Appendix: Full Cov[$\hat{w}$] derivation

INFO 521 · Module 3 — supplement

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Where we are

From the main deck: $\hat{\mathbf{w}}$ is unbiased ($\mathbb{E}[\hat{\mathbf{w}}] = \mathbf{w}$), and the parameter covariance is

$$\mathrm{Cov}[\hat{\mathbf{w}}] \;=\; \sigma^2 \left(\mathbf{X}^\top \mathbf{X}\right)^{-1}.$$

This appendix derives that result in full. Only $\mathbf{y}$ is random; $\mathbf{X}$ and $\mathbf{w}$ are fixed, and $\mathbb{E}[\mathbf{y}] = \mathbf{X}\mathbf{w}$, $\mathrm{Cov}[\mathbf{y}] = \sigma^2\mathbf{I}$.

Set up the second moment

Start from the covariance definition, using $\mathbb{E}[\hat{\mathbf{w}}] = \mathbf{w}$:

$$\mathrm{Cov}[\hat{\mathbf{w}}] \; = \; \mathbb{E}\left[\hat{\mathbf{w}}\hat{\mathbf{w}}^\top\right] - \mathbf{w}\mathbf{w}^\top$$

Substitute $\hat{\mathbf{w}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$ and pull the constants out of the expectation:

$$\mathbb{E}\left[\hat{\mathbf{w}}\hat{\mathbf{w}}^\top\right] = \left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\,\mathbb{E}\left[\mathbf{y}\mathbf{y}^\top\right]\,\mathbf{X}\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}$$

The data second moment, from $\mathrm{Cov}[\mathbf{y}] = \mathbb{E}[\mathbf{y}\mathbf{y}^\top] - \mathbb{E}[\mathbf{y}]\mathbb{E}[\mathbf{y}]^\top$:

$$\mathbb{E}\left[\mathbf{y}\mathbf{y}^\top\right] = \mathbf{X}\mathbf{w}\mathbf{w}^\top\mathbf{X}^\top + \sigma^2\mathbf{I}$$

Collect the terms

Substitute and use $(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X} = \mathbf{I}$ on each piece:

$$\mathbb{E}\left[\hat{\mathbf{w}}\hat{\mathbf{w}}^\top\right] = \mathbf{w}\mathbf{w}^\top + \sigma^2\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}$$

The $\mathbf{w}\mathbf{w}^\top$ cancels against the $-\mathbf{w}\mathbf{w}^\top$ from the definition:

$$\boxed{\mathrm{Cov}[\hat{\mathbf{w}}] \;=\; \sigma^2\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}}$$

And this is exactly the inverse Fisher information: from $-\dfrac{\partial^2\ell}{\partial\mathbf{w}\,\partial\mathbf{w}^\top} = \frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X} = \mathcal{I}$ (with ℓ the log-likelihood from the main deck), we get $\mathrm{Cov}[\hat{\mathbf{w}}] = \mathcal{I}^{-1}$.